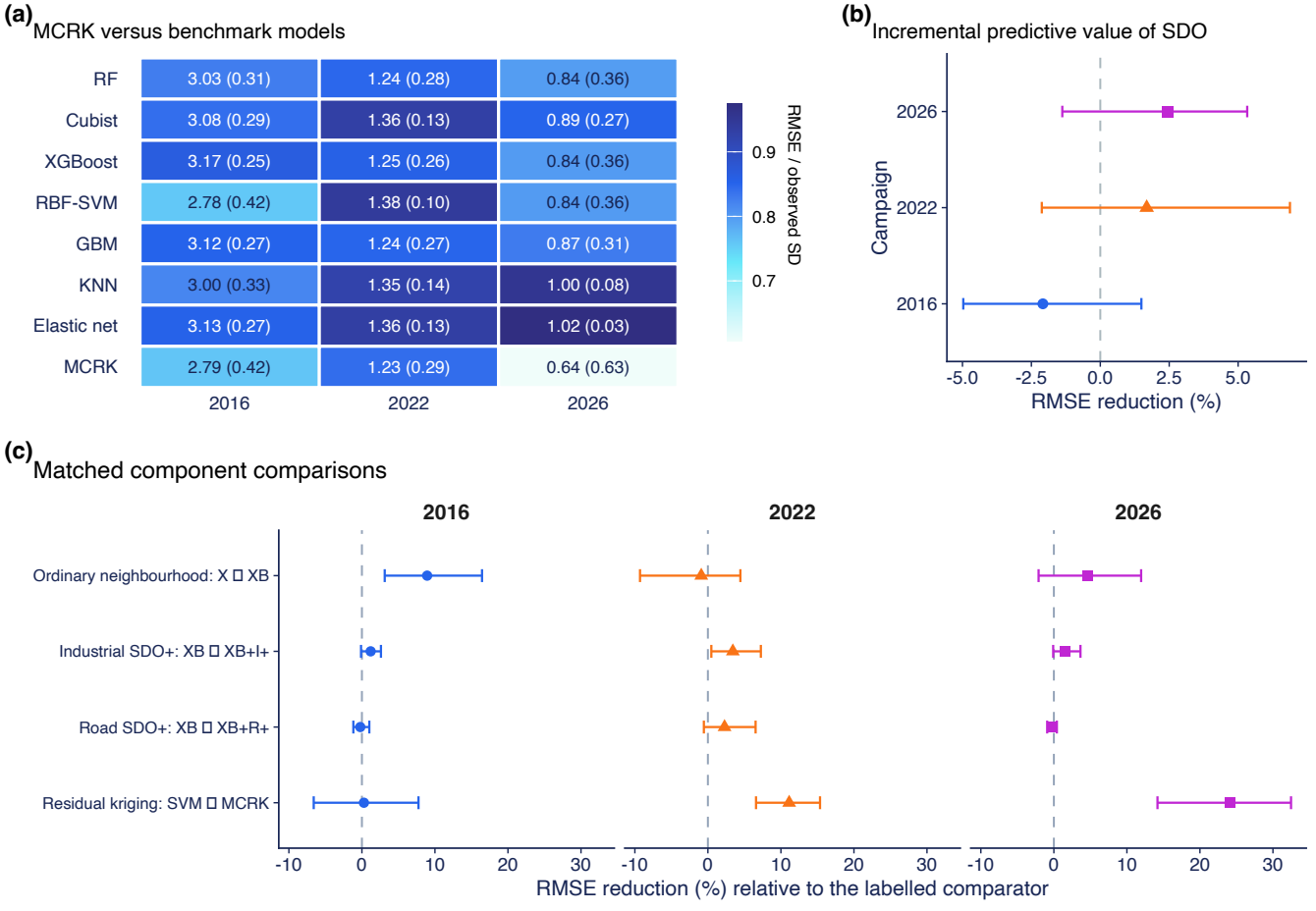
(a)
MCRK versus benchmark models
RF
Cubist
XGBoost
RBF-SVM
GBM
KNN
Elastic net
MCRK
3.03 (0.31)
1.24 (0.28)
0.84 (0.36)
3.08 (0.29)
1.36 (0.13)
0.89 (0.27)
3.17 (0.25)
1.25 (0.26)
0.84 (0.36)
2.78 (0.42)
1.38 (0.10)
0.84 (0.36)
3.12 (0.27)
1.24 (0.27)
0.87 (0.31)
3.00 (0.33)
1.35 (0.14)
1.00 (0.08)
3.13 (0.27)
1.36 (0.13)
1.02 (0.03)
2.79 (0.42)
1.23 (0.29)
0.64 (0.63)
2016
2022
2026
RMSE / observed SD
0.9
0.8
0.7
(b)
Incremental predictive value of SDO
Campaign
2026
2022
2016
-5.0
-2.5
0.0
2.5
5.0
RMSE reduction (%)
(c)
Matched component comparisons
2016
2022
2026
Ordinary neighbourhood: X □ XB
Industrial SDO+: XB □ XB+I+
Road SDO+: XB □ XB+R+
Residual kriging: SVM □ MCRK
-10
0
10
20
30
RMSE reduction (%) relative to the labelled comparator